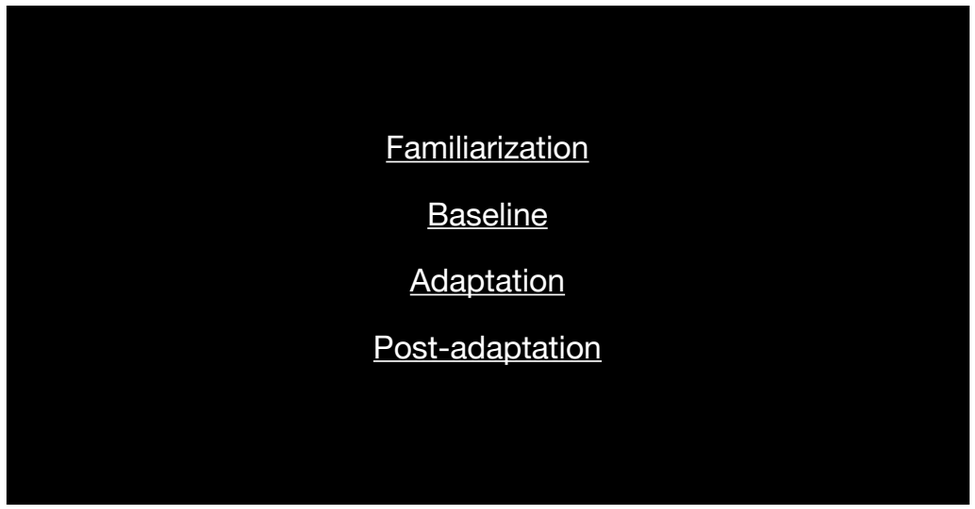
起始畫面



每一個可以點進去 然後想要點進去之後有先我命名儲存檔案的block

1. Familiarization
- Familiarization ×16

Total = 16 trials

2. Baseline
- Veridical ×34
- No Vision ×6
- Report Training ×40

Total = 80 trials

3. Adaptation

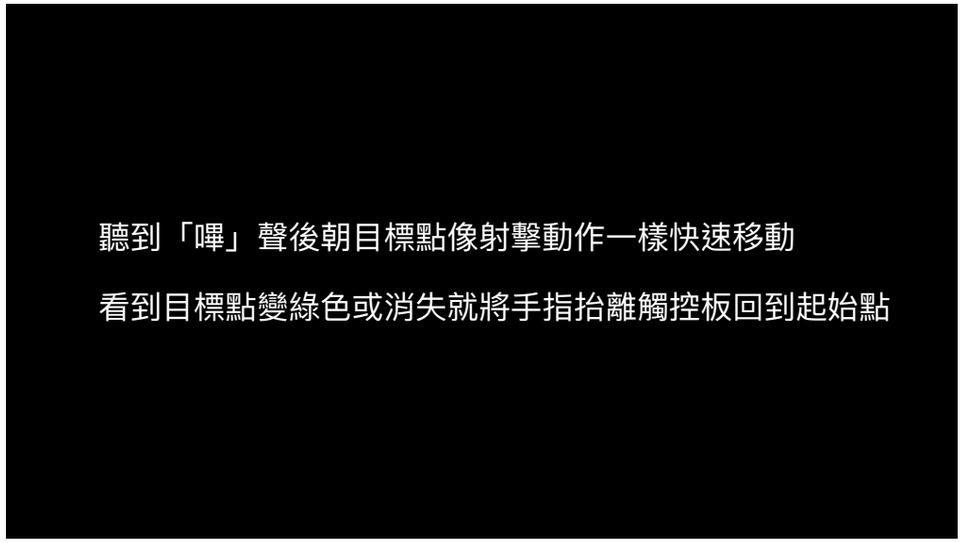
- Regular Rotated ×40
- Aim Report ×4
- Return-to-baseline / Exclusion ×4

Each cycle = 48 trials

4. Post-adaptation
- Exclusion / Return-to-baseline ×16
- Re-adaptation / Refresh Adaptation ×16
- Inclusion / Use-learned-adjustment ×16

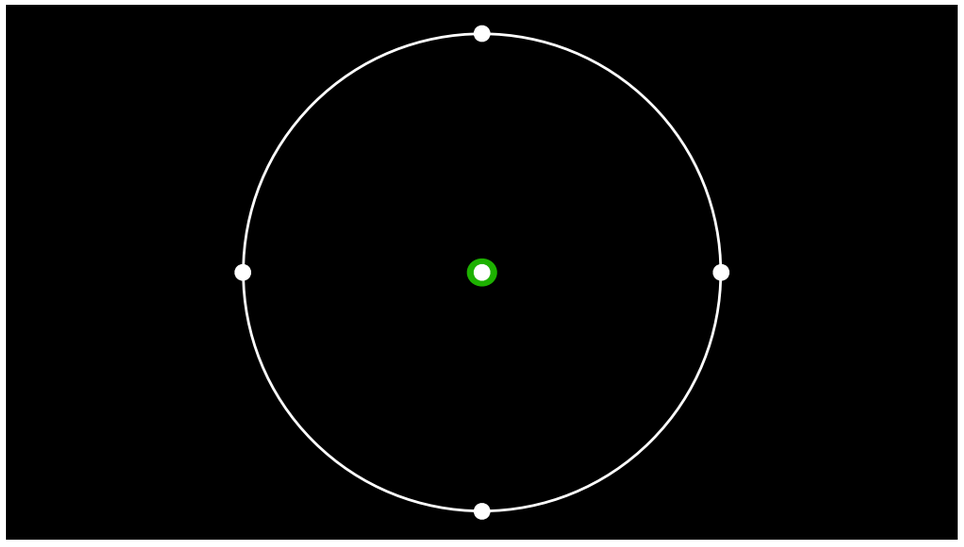
Total = 48 trials

Familiarization



Total: 16 trials

Familiarization-出發前



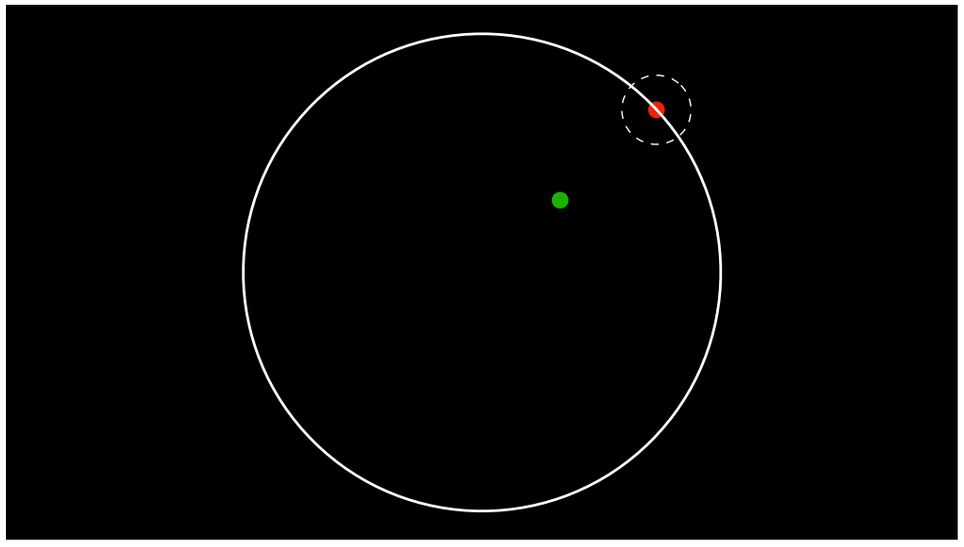
要在綠色的地方等500ms才會有target 出現 然後home的綠色變白色 target 出現是紅色的（這邊擺八個點是要隨機出現在這些位置 受試者不會看到有4點 只會看到目標點）

```
Layout:
  screen_home_position_px （中心點）:
    x: 960
    y: 540
  target_layout:
    type: circle
    n_targets: 4
    target_angles_deg: [0, 90, 180, 270]
    pseudorandom_order: true
  home_to_target_distance:
    cm: 7.0
    px: 391
  workspace_boundary_radius_px: 391
  cursor_radius:
    mm: 0.5
    px: 3
  home_radius:
    mm: 2.5
    px: 14
```

```
  target_radius:
    mm: 5.0
    px: 28
timing:
  inter_trial_interval_ms: 2000
  home_hold_ms: 500
  target_timeout_ms: 2000
  end_feedback_duration_ms: 300
  report_timeout_ms: null

go_signal:
  mode: audio_beep
  audio_file: go_beep
  home_color_change_on_go: true
  home_color_before_go: green
```

Familiarization-移動中



移動過程cursor、target都看得到 但最外圈白色跟目標外圈虛線是看不到的 只是用來判定這個trial 是否結束

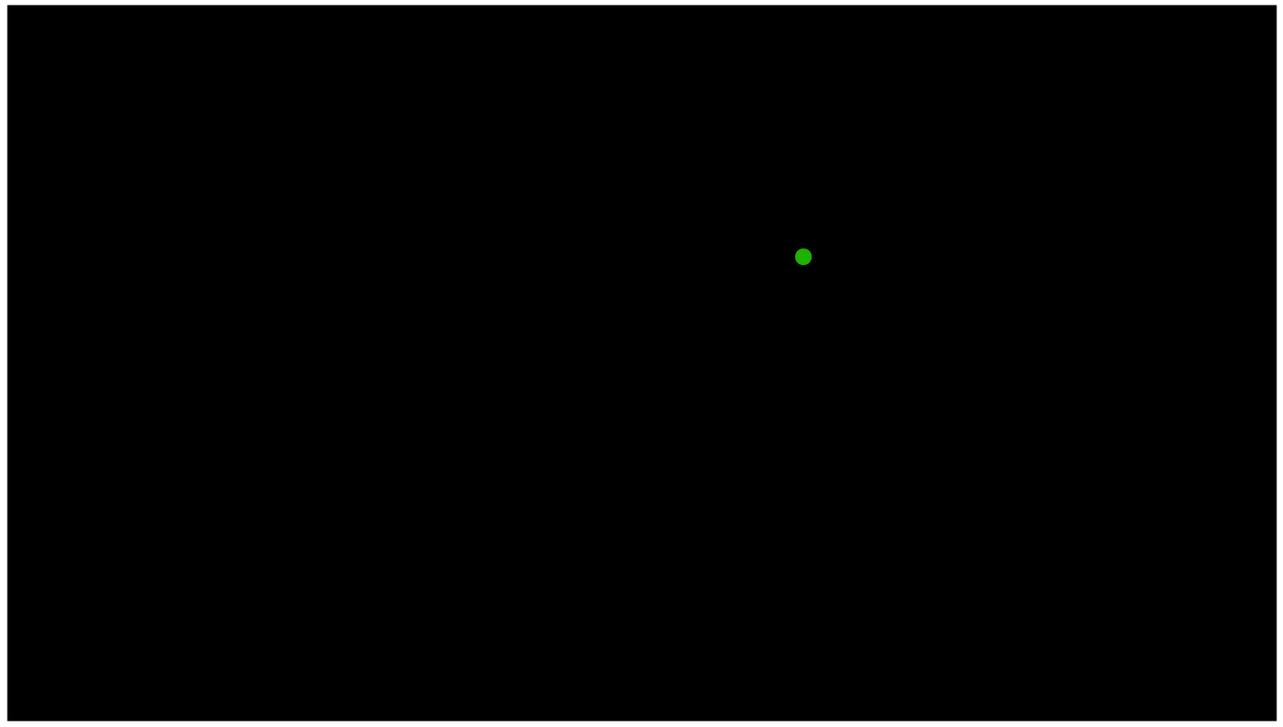
success是cursor碰到target外面那圈虛線

fail 是超過時間（如上頁規定）或是碰到白色實線

screen:

- target: visible
- cursor: visible
- landmarks: hidden

目標外的7cm 有一圈



```
end_condition:
    - hit_target
    - exceed_workspace_boundary
    - target_timeout
feedback:
  speed_band_ms:
    good_min: 400
    good_max: 600
  target_color_feedback:
          Hit situation:
                    good_speed: green
                     too_fast: yellow
                     too_slow: blue
           exceed_workspace_boundary/target_timeout:
                    target disappear
  auditory_feedback:
    hit_sound: hit_bell.wav
    miss_sound: miss_buzz.wav
Data_collection:
          - movement angle in all trials: home to movement midpoint
```

-

Baseline- section start

聽到「嗶」聲後朝目標點像射擊動作一樣快速移動

看到目標點變綠色或消失就將手指抬離觸控板回到起始點

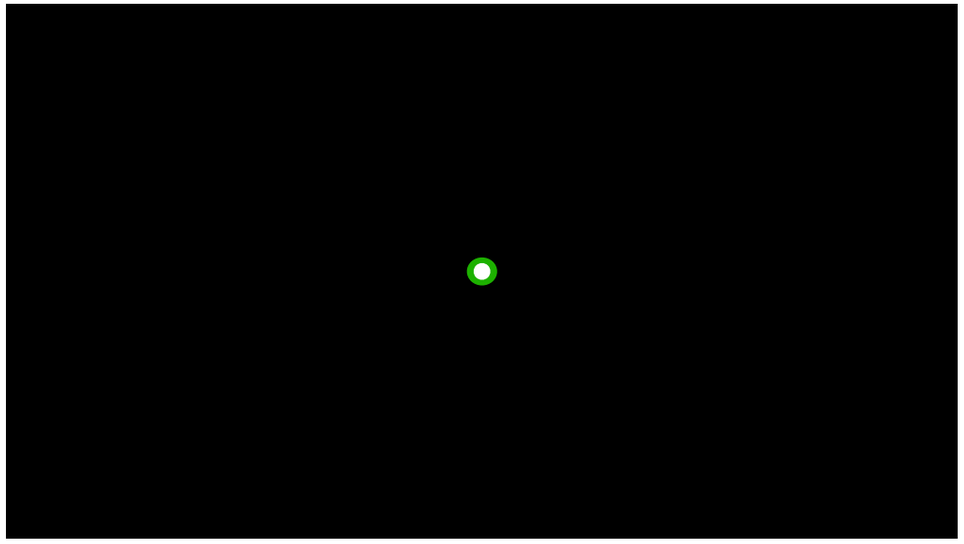
有些回合會沒有游標，照原本的移動方式動就好

有些回合需要您講出要向哪個數字移動後再執行

Total: 80 trials

- 前34 veridical+6 no vision：要pseudorandomly interspersed
- 後40trial report training

Baseline-veridical trial （起始）



Start layout:
- Home visible at screen center, green
- Cursor visible at home
- Target initially hidden
- Landmarks hidden
- Workspace boundary hidden

Go signal:
- Participant stays within home for 500 ms
- One target appears at one of the 4 possible target locations
- Target color = red
- Audio beep
- Home disappears at movement onset

Movement display:
- Target visible
- Cursor visible continuously during movement
- Home hidden
- Landmarks hidden

Cursor mapping:
- Veridical mapping
- rotation_deg = 0°

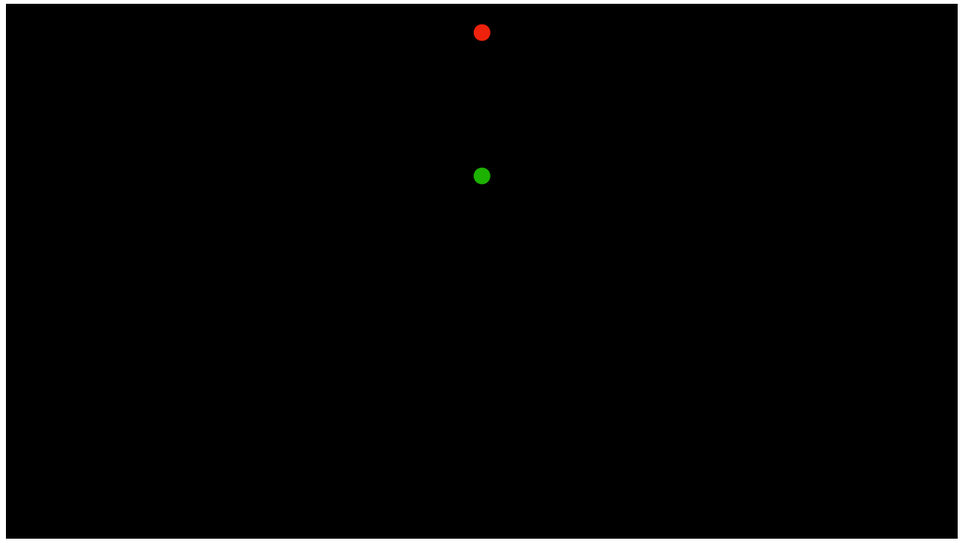
End condition:
- Display cursor enters target zone
- Raw movement crosses workspace boundary (391 px from home)
- Target timeout = 2000 ms

Feedback:
- Hit + movement time 400–600 ms → target green + hit bell
- Hit + movement time <400 ms → target yellow
- Hit + movement time >600 ms → target blue
- Workspace boundary exceeded / timeout → target disappears + miss buzz

Data to record:
- trial_type = baseline_veridical
- trial_index
- target_id
- target_angle_deg
- target_x, target_y
- rotation_deg = 0
- target_onset_timestamp
- go_signal_timestamp
- movement_onset_timestamp
- movement_end_timestamp
- raw_x, raw_y, sample_timestamp
- display_x, display_y
- hit_target
- boundary_crossed
- timeout
- movement_time_ms

Derived offline:
- movement_angle_at_50pct_reach

Baseline-veridical trial （移動）



Start layout:
- Home visible at screen center, green
- Cursor visible at home
- Target initially hidden
- Landmarks hidden
- Workspace boundary hidden

Go signal:
- Participant stays within home for 500 ms
- One target appears at one of the 4 possible target locations
- Target color = red
- Audio beep
- Home disappears at movement onset

Movement display:
- Target visible
- Cursor visible continuously during movement
- Home hidden
- Landmarks hidden

Cursor mapping:
- Veridical mapping
- rotation_deg = 0°

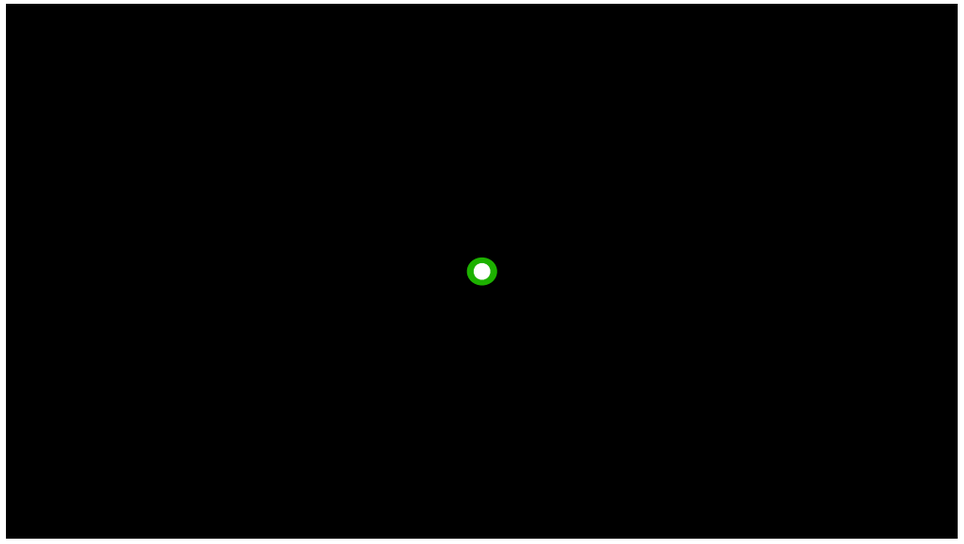
End condition:
- Display cursor enters target zone
- Raw movement crosses workspace boundary (391 px from home)
- Target timeout = 2000 ms

Feedback:
- Hit + movement time 400–600 ms → target green + hit bell
- Hit + movement time <400 ms → target yellow
- Hit + movement time >600 ms → target blue
- Workspace boundary exceeded / timeout → target disappears + miss buzz

Data to record:
- trial_type = baseline_veridical
- trial_index
- target_id
- target_angle_deg
- target_x, target_y
- rotation_deg = 0
- target_onset_timestamp
- go_signal_timestamp
- movement_onset_timestamp
- movement_end_timestamp
- raw_x, raw_y, sample_timestamp
- display_x, display_y
- hit_target
- boundary_crossed
- timeout
- movement_time_ms

Derived offline:
- movement_angle_at_50pct_reach

Baseline-no vision trial （起始）



Start layout:
- Home visible at screen center, green
- Cursor visible at home
- Target initially hidden
- Landmarks hidden
- Workspace boundary hidden

Go signal:
- Participant stays within home for 500 ms
- One target appears
- Target color = red
- Audio beep
- Home disappears at movement onset
- Cursor disappears at movement onset

Movement display:
- Target visible
- Home hidden
- Cursor hidden
- Landmarks hidden

Cursor mapping:
- No visual movement feedback
- Raw pointer movement continues to be recorded

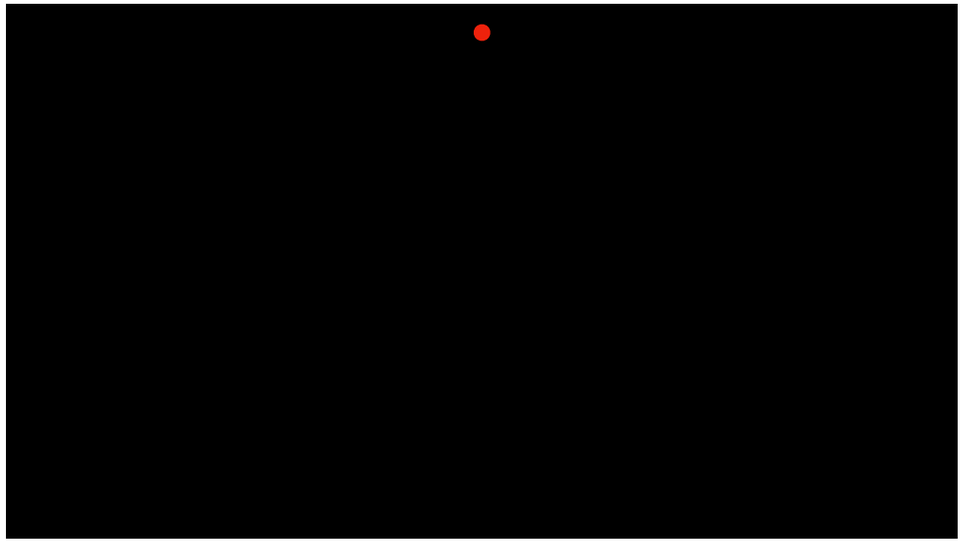
End condition:
- Raw movement enters target zone
- Raw movement crosses workspace boundary (391 px)
- Target timeout = 2000 ms

Feedback:
- Hit + movement time 400–600 ms → target green + hit bell
- Hit + movement time <400 ms → target yellow
- Hit + movement time >600 ms → target blue
- Workspace boundary exceeded / timeout → target disappears + miss buzz

Data to record:
- trial_type = baseline_no_vision
- trial_index
- target_id
- target_angle_deg
- target_x, target_y
- target_onset_timestamp
- go_signal_timestamp
- movement_onset_timestamp
- movement_end_timestamp
- raw_x, raw_y, sample_timestamp
- cursor_visible_during_movement = false
- hit_target
- boundary_crossed
- timeout
- movement_time_ms

Derived offline:
- movement_angle_at_50pct_reach

Baseline-no vision trial （移動）



移動過程看不到cursor

Start layout:
- Home visible at screen center, green
- Cursor visible at home
- Target initially hidden
- Landmarks hidden
- Workspace boundary hidden

Go signal:
- Participant stays within home for 500 ms
- One target appears
- Target color = red
- Audio beep
- Home disappears at movement onset
- Cursor disappears at movement onset

Movement display:
- Target visible
- Home hidden
- **Cursor hidden**

- Landmarks hidden

Cursor mapping:
- No visual movement feedback
- Raw pointer movement continues to be recorded

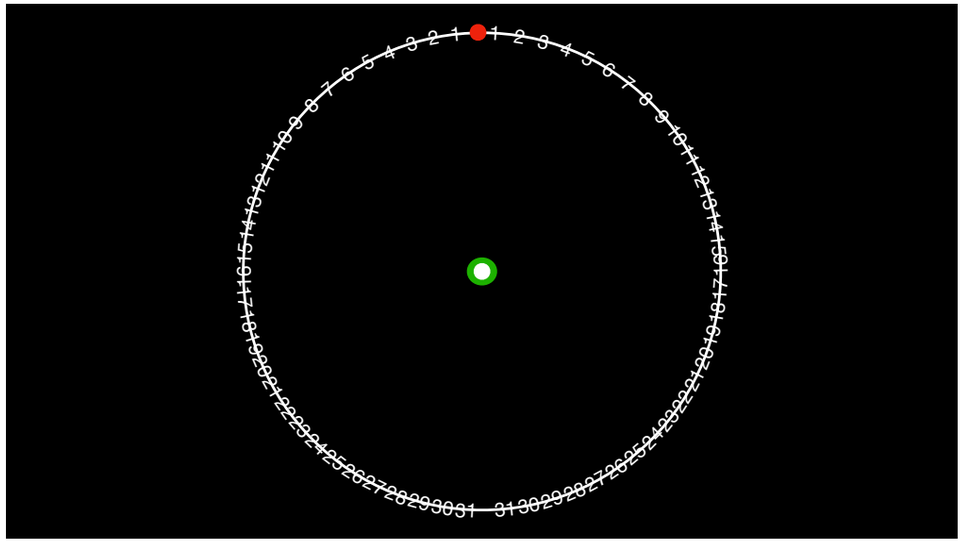
End condition:
- Raw movement enters target zone
- Raw movement crosses workspace boundary (391 px)
- Target timeout = 2000 ms

Feedback:
- Hit + movement time 400–600 ms → target green + hit bell
- Hit + movement time <400 ms → target yellow
- Hit + movement time >600 ms → target blue
- Workspace boundary exceeded / timeout → target disappears + miss buzz

Data to record:
- trial_type = baseline_no_vision
- trial_index
- target_id
- target_angle_deg
- target_x, target_y
- target_onset_timestamp
- go_signal_timestamp
- movement_onset_timestamp
- movement_end_timestamp
- raw_x, raw_y, sample_timestamp
- cursor_visible_during_movement = false
- hit_target
- boundary_crossed
- timeout
- movement_time_ms

Derived offline:
- movement_angle_at_50pct_reach

Baseline-report training trial （起始）



Baseline – Report-training Trial

Start layout:
- Home visible at screen center, green
- Cursor visible at home
- Target initially hidden
- Numbered landmarks initially hidden
- Workspace boundary hidden

Target selection:
- One target is selected pseudorandomly from the 4 possible target locations
- target_angles_deg = [0, 90, 180, 270]
- Target direction is independent of the training rotation

Report phase:
- Participant stays within home for 500 ms
- One target appears, red
- Numbered landmarks appear
- The entire landmark ring rotates with the current target direction
- Target direction serves as the 0° aiming reference

- Therefore, the same landmark numbers always represent the same angular offsets relative to the target
- Home remains visible during the report phase
- Participant verbally reports the intended aiming number / direction
- No movement is allowed yet
- No report timeout

Experimenter confirmation:
- Wait for experimenter to press SPACE
- SPACE confirms that the verbal report has been recorded
- Record the reported number before continuing

Go signal:
- After SPACE press:
  - Audio beep
  - Movement is enabled
- Home disappears at movement onset

Movement display:
- Target visible
- Cursor visible continuously during movement
- Numbered landmarks visible
- Home hidden
- Workspace boundary hidden

Cursor mapping:
- Trial-specific variable visuomotor rotation
- training_rotation_deg varies approximately from -30° to +30°
- The rotation changes from trial to trial
- training_rotation_deg affects cursor feedback only
- It does NOT change target position or landmark reference direction
- No stable rotation is used during report training

End condition:
- Display cursor enters target zone
- Raw movement crosses workspace boundary (391 px from home)
- Target timeout = 2000 ms after SPACE / movement is enabled

Feedback:
- Hit + movement time 400–600 ms → target turns green + hit bell
- Hit + movement time <400 ms → target turns yellow
- Hit + movement time >600 ms → target turns blue
- Workspace boundary exceeded / timeout → target disappears + miss buzz

Data to record:
- trial_type = baseline_report_training
- trial_index

- target_id
- target_angle_deg
- target_x
- target_y

- training_rotation_deg

- reported_number
- reported_side
- reported_angle_deg
- report_registered_timestamp
- experimenter_space_timestamp

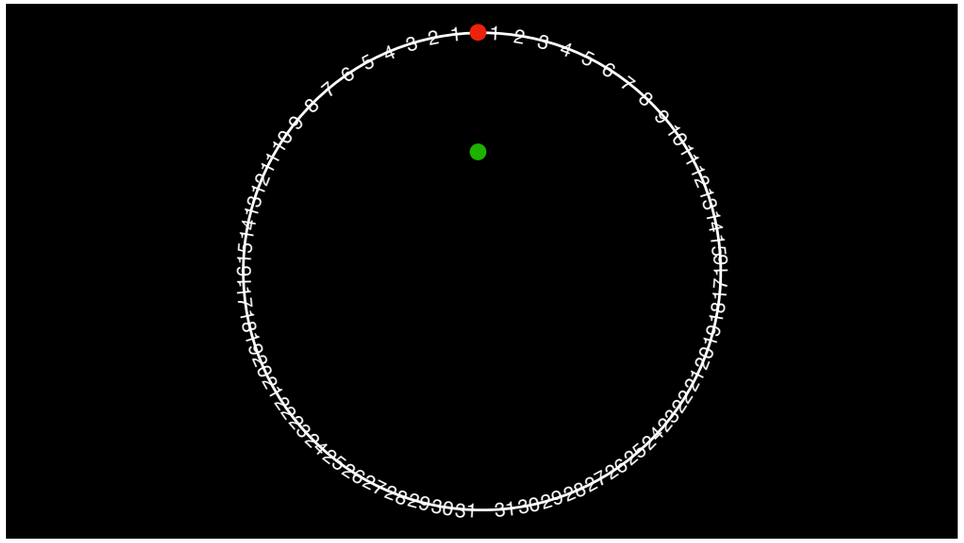
- target_onset_timestamp
- go_signal_timestamp
- movement_onset_timestamp
- movement_end_timestamp

- raw_x
- raw_y
- sample_timestamp

- display_x
- display_y

- hit_target
- boundary_crossed
- timeout
- movement_time_ms

Baseline-report training trial （移動）



Baseline – Report-training Trial

Start layout:
- Home visible at screen center, green
- Cursor visible at home
- Target initially hidden
- Numbered landmarks initially hidden
- Workspace boundary hidden

Target selection:
- One target is selected pseudorandomly from the 4 possible target locations
- target_angles_deg = [0, 90, 180, 270]
- Target direction is independent of the training rotation

Report phase:
- Participant stays within home for 500 ms
- One target appears, red
- Numbered landmarks appear
- The entire landmark ring rotates with the current target direction
- Target direction serves as the 0° aiming reference

- Therefore, the same landmark numbers always represent the same angular offsets relative to the target
- Home remains visible during the report phase
- Participant verbally reports the intended aiming number / direction
- No movement is allowed yet
- No report timeout

Experimenter confirmation:
- Wait for experimenter to press SPACE
- SPACE confirms that the verbal report has been recorded
- Record the reported number before continuing

Go signal:
- After SPACE press:
  - Audio beep
  - Movement is enabled
- Home disappears at movement onset

Movement display:
- Target visible
- Cursor visible continuously during movement
- Numbered landmarks visible
- Home hidden
- Workspace boundary hidden

Cursor mapping:
- Trial-specific variable visuomotor rotation
- training_rotation_deg varies approximately from -30° to +30°
- The rotation changes from trial to trial
- training_rotation_deg affects cursor feedback only
- It does NOT change target position or landmark reference direction
- No stable rotation is used during report training

End condition:
- Display cursor enters target zone
- Raw movement crosses workspace boundary (391 px from home)
- Target timeout = 2000 ms after SPACE / movement is enabled

Feedback:
- Hit + movement time 400–600 ms → target turns green + hit bell
- Hit + movement time <400 ms → target turns yellow
- Hit + movement time >600 ms → target turns blue
- Workspace boundary exceeded / timeout → target disappears + miss buzz

Data to record:
- trial_type = baseline_report_training
- trial_index

- target_id
- target_angle_deg
- target_x
- target_y

- training_rotation_deg

- reported_number
- reported_side
- reported_angle_deg
- report_registered_timestamp
- experimenter_space_timestamp

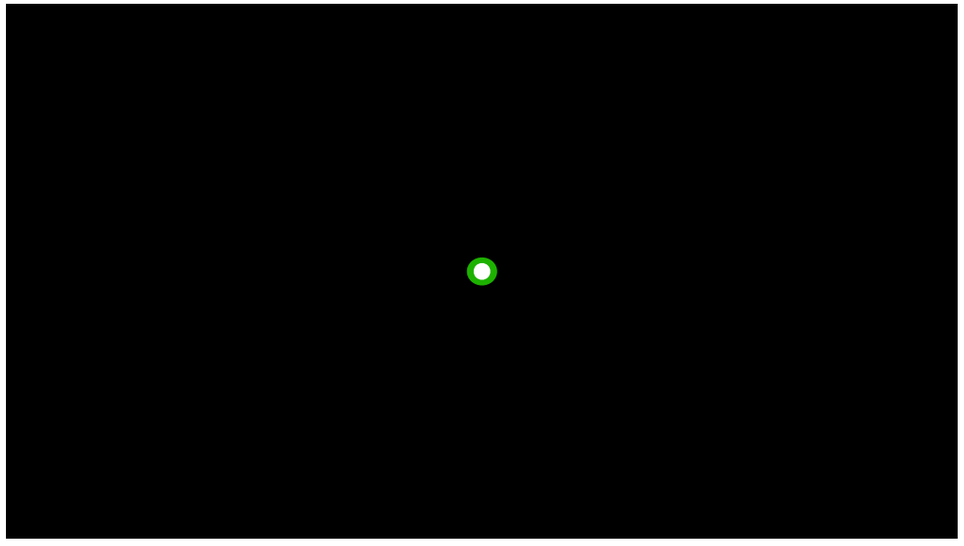
- target_onset_timestamp
- go_signal_timestamp
- movement_onset_timestamp
- movement_end_timestamp

- raw_x
- raw_y
- sample_timestamp

- display_x
- display_y

- hit_target
- boundary_crossed
- timeout
- movement_time_ms

Adaptation- rotated trial (起始)



Start layout:
- Home visible at screen center, green
- Cursor visible at home
- Target initially hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Target selection:
- One target is selected pseudorandomly from the 4 possible target locations
- target_angles_deg = [0, 90, 180, 270]
- Only the selected target is visible to the participant

Go signal:
- Participant stays within home for 500 ms
- One target appears, red
- Audio beep
- Movement is enabled
- Home disappears at movement onset

Movement display:

- Target remains visible
- Rotated cursor visible continuously during movement
- Home hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Cursor mapping:
- Fixed visuomotor rotation
- rotation_deg = 60
- Same rotation magnitude and direction throughout the adaptation phase
- Rotation is applied to the displayed cursor only; raw movement is recorded separately

End condition:
- Display cursor enters target zone
- Raw movement crosses workspace boundary (391 px from home)
- Target timeout = 2000 ms

Feedback:
- Hit + movement time 400–600 ms
  → target turns green + hit bell
- Hit + movement time <400 ms
  → target turns yellow
- Hit + movement time >600 ms
  → target turns blue
- Workspace boundary exceeded / timeout
  → target disappears + miss buzz

Data to record:
- trial_type = adaptation_regular
- adaptation_block
- trial_index

- target_id
- target_angle_deg
- target_x
- target_y

- rotation_deg

- target_onset_timestamp
- go_signal_timestamp
- movement_onset_timestamp

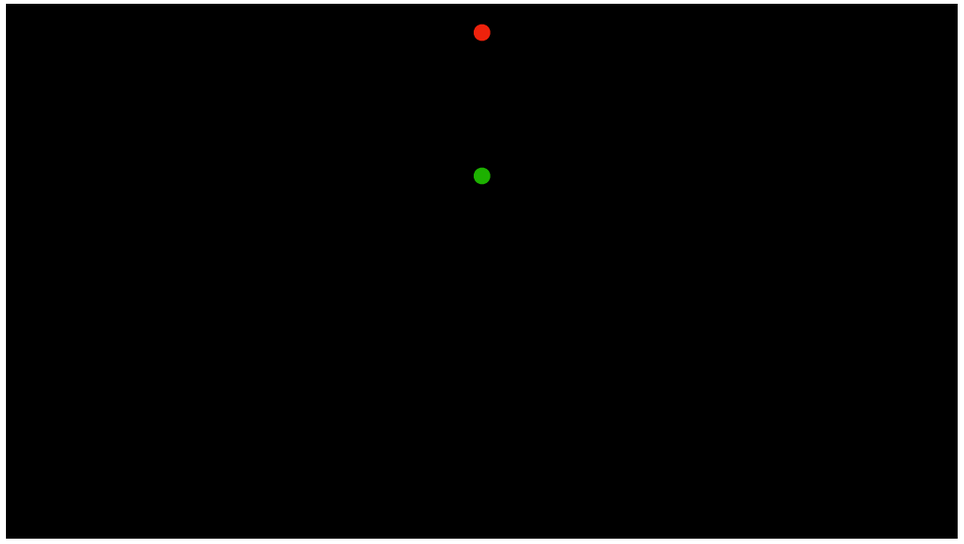
- movement_end_timestamp

- raw_x
- raw_y
- sample_timestamp
  (record every available input event; do not restrict recording to display frames)

- display_x
- display_y

- hit_target
- boundary_crossed
- timeout

Adaptation- rotated trial (過程)



Start layout:
- Home visible at screen center, green
- Cursor visible at home
- Target initially hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Target selection:
- One target is selected pseudorandomly from the 4 possible target locations
- target_angles_deg = [0, 90, 180, 270]
- Only the selected target is visible to the participant

Go signal:
- Participant stays within home for 500 ms
- One target appears, red
- Audio beep
- Movement is enabled
- Home disappears at movement onset

Movement display:

- Target remains visible
- Rotated cursor visible continuously during movement
- Home hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Cursor mapping:
- Fixed visuomotor rotation
- rotation_deg = 60
- Same rotation magnitude and direction throughout the adaptation phase
- Rotation is applied to the displayed cursor only; raw movement is recorded separately

End condition:
- Display cursor enters target zone
- Raw movement crosses workspace boundary (391 px from home)
- Target timeout = 2000 ms

Feedback:
- Hit + movement time 400–600 ms
  → target turns green + hit bell
- Hit + movement time <400 ms
  → target turns yellow
- Hit + movement time >600 ms
  → target turns blue
- Workspace boundary exceeded / timeout
  → target disappears + miss buzz

Data to record:
- trial_type = adaptation_regular
- adaptation_block
- trial_index

- target_id
- target_angle_deg
- target_x
- target_y

- rotation_deg

- target_onset_timestamp
- go_signal_timestamp
- movement_onset_timestamp

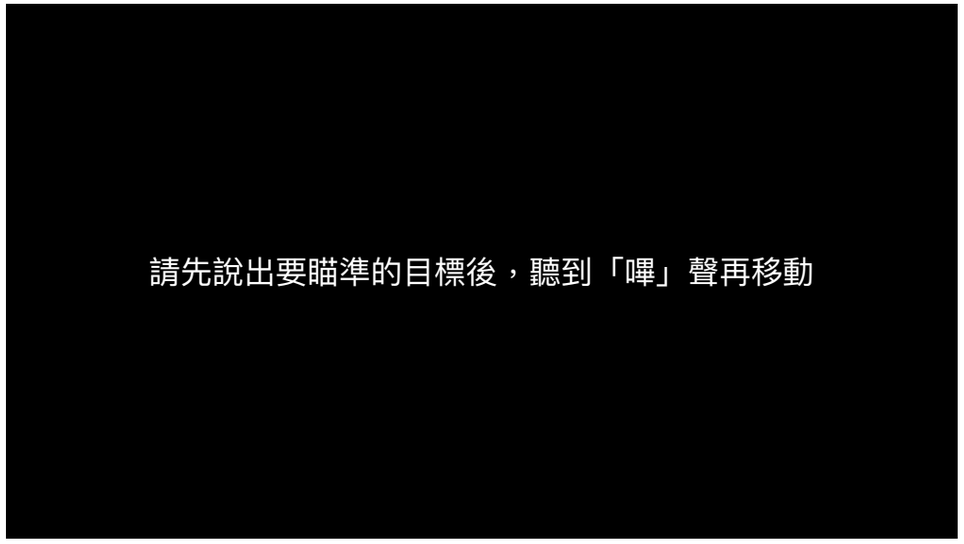
- movement_end_timestamp

- raw_x
- raw_y
- sample_timestamp
  (record every available input event; do not restrict recording to display frames)

- display_x
- display_y

- hit_target
- boundary_crossed
- timeout

Adaptation-report trial section start



Start layout:
- Home visible at screen center, green
- Cursor visible at home
- Target initially hidden
- Numbered landmarks initially hidden
- Workspace boundary hidden

Target selection:
- One target is selected pseudorandomly from the 4 possible target locations
- target_angles_deg = [0, 90, 180, 270]
- Only the selected target is visible to the participant

Report phase:
- Participant stays within home for 500 ms
- One target appears, red
- Numbered landmarks appear
- The landmark ring rotates with the current target direction
- Target direction serves as the 0° aiming reference
- Home remains visible during the verbal report
- Participant verbally reports the intended aiming number / direction

- No movement is allowed yet
- No report timeout

Experimenter confirmation:
- Wait for experimenter to press SPACE
- SPACE confirms that the verbal report has been recorded

Go signal:
- After SPACE press:
  - Audio beep
  - Movement is enabled
- Home disappears at movement onset

Movement display:
- Target remains visible
- Numbered landmarks remain visible
- Rotated cursor visible continuously during movement
- Home hidden
- Workspace boundary hidden

Cursor mapping:
- Fixed visuomotor rotation
- Cursor feedback is rotated 60° clockwise relative to the raw hand / touchpad movement
- The target itself does not rotate or move because of the 60° perturbation
- The same 60° clockwise rotation is used throughout the adaptation phase
- Raw movement is recorded separately from displayed cursor position

End condition:
- Display cursor enters target zone
- Raw movement crosses workspace boundary (391 px from home)
- Target timeout = 2000 ms after SPACE / movement is enabled

Feedback:
- Hit + movement time 400–600 ms
  → target turns green + hit bell
- Hit + movement time <400 ms
  → target turns yellow
- Hit + movement time >600 ms
  → target turns blue
- Workspace boundary exceeded / timeout
  → target disappears + miss buzz

Data to record:
- trial_type = adaptation_report
- adaptation_block
- probe_number
- trial_index

- target_id
- target_angle_deg
- target_x
- target_y

- rotation_direction = clockwise
- rotation_deg = 60

- reported_number
- reported_side
- reported_angle_deg
- report_registered_timestamp
- experimenter_space_timestamp

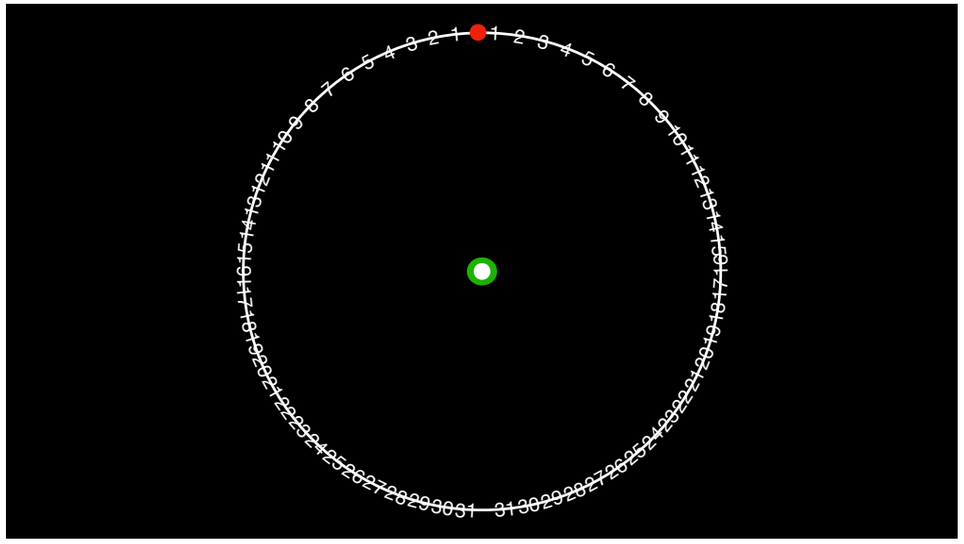
- target_onset_timestamp
- go_signal_timestamp
- movement_onset_timestamp
- movement_end_timestamp

- raw_x
- raw_y
- sample_timestamp
  (record every available input event; do not restrict recording to display frames)

- display_x
- display_y

- hit_target
- boundary_crossed
- timeout

Adaptation-report trial （起始）



Start layout:
- Home visible at screen center, green
- Cursor visible at home
- Target initially hidden
- Numbered landmarks initially hidden
- Workspace boundary hidden

Target selection:
- One target is selected pseudorandomly from the 4 possible target locations
- target_angles_deg = [0, 90, 180, 270]
- Only the selected target is visible to the participant

Report phase:
- Participant stays within home for 500 ms
- One target appears, red
- Numbered landmarks appear
- The landmark ring rotates with the current target direction
- Target direction serves as the 0° aiming reference
- Home remains visible during the verbal report
- Participant verbally reports the intended aiming number / direction

- No movement is allowed yet
- No report timeout

Experimenter confirmation:
- Wait for experimenter to press SPACE
- SPACE confirms that the verbal report has been recorded

Go signal:
- After SPACE press:
  - Audio beep
  - Movement is enabled
- Home disappears at movement onset

Movement display:
- Target remains visible
- Numbered landmarks remain visible
- Rotated cursor visible continuously during movement
- Home hidden
- Workspace boundary hidden

Cursor mapping:
- Fixed visuomotor rotation
- Cursor feedback is rotated 60° clockwise relative to the raw hand / touchpad movement
- The target itself does not rotate or move because of the 60° perturbation
- The same 60° clockwise rotation is used throughout the adaptation phase
- Raw movement is recorded separately from displayed cursor position

End condition:
- Display cursor enters target zone
- Raw movement crosses workspace boundary (391 px from home)
- Target timeout = 2000 ms after SPACE / movement is enabled

Feedback:
- Hit + movement time 400–600 ms
  → target turns green + hit bell
- Hit + movement time <400 ms
  → target turns yellow
- Hit + movement time >600 ms
  → target turns blue
- Workspace boundary exceeded / timeout
  → target disappears + miss buzz

Data to record:
- trial_type = adaptation_report
- adaptation_block
- probe_number
- trial_index

- target_id
- target_angle_deg
- target_x
- target_y

- rotation_direction = clockwise
- rotation_deg = 60

- reported_number
- reported_side
- reported_angle_deg
- report_registered_timestamp
- experimenter_space_timestamp

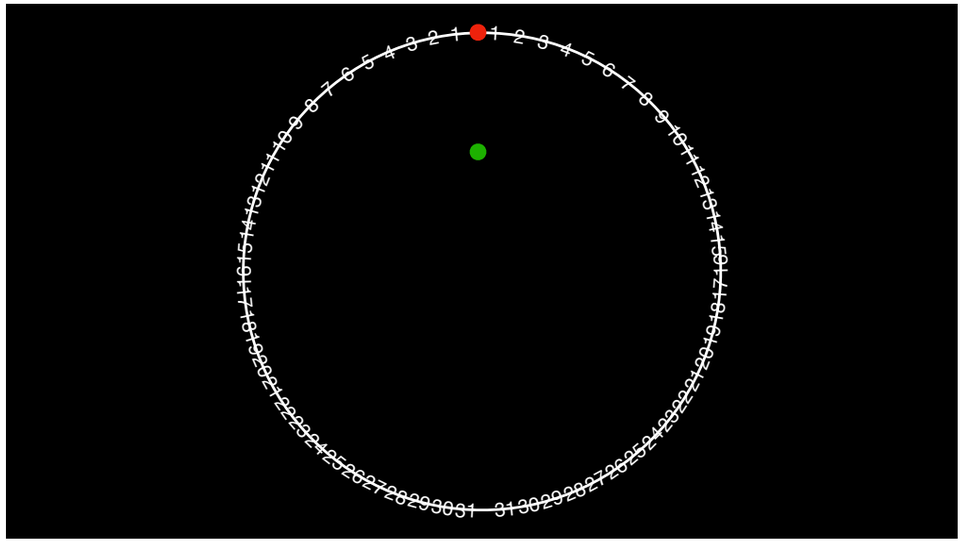
- target_onset_timestamp
- go_signal_timestamp
- movement_onset_timestamp
- movement_end_timestamp

- raw_x
- raw_y
- sample_timestamp
  (record every available input event; do not restrict recording to display frames)

- display_x
- display_y

- hit_target
- boundary_crossed
- timeout

Adaptation-report trial （移動）



Start layout:
- Home visible at screen center, green
- Cursor visible at home
- Target initially hidden
- Numbered landmarks initially hidden
- Workspace boundary hidden

Target selection:
- One target is selected pseudorandomly from the 4 possible target locations
- target_angles_deg = [0, 90, 180, 270]
- Only the selected target is visible to the participant

Report phase:
- Participant stays within home for 500 ms
- One target appears, red
- Numbered landmarks appear
- The landmark ring rotates with the current target direction
- Target direction serves as the 0° aiming reference
- Home remains visible during the verbal report
- Participant verbally reports the intended aiming number / direction

- No movement is allowed yet
- No report timeout

Experimenter confirmation:
- Wait for experimenter to press SPACE
- SPACE confirms that the verbal report has been recorded

Go signal:
- After SPACE press:
  - Audio beep
  - Movement is enabled
- Home disappears at movement onset

Movement display:
- Target remains visible
- Numbered landmarks remain visible
- Rotated cursor visible during movement
- Home hidden
- Workspace boundary hidden

Cursor mapping:
- Fixed visuomotor rotation
- Cursor feedback is rotated 60° clockwise relative to the raw hand / touchpad movement
- The target itself does not rotate or move because of the 60° perturbation
- The same 60° clockwise rotation is used throughout the adaptation phase
- Raw movement is recorded separately from displayed cursor position

End condition:
- Display cursor enters target zone
- Raw movement crosses workspace boundary (391 px from home)
- Target timeout = 2000 ms after SPACE / movement is enabled

Feedback:
- Hit + movement time 400–600 ms
  → target turns green + hit bell
- Hit + movement time <400 ms
  → target turns yellow
- Hit + movement time >600 ms
  → target turns blue
- Workspace boundary exceeded / timeout
  → target disappears + miss buzz

Data to record:
- trial_type = adaptation_report
- adaptation_block
- probe_number
- trial_index

- target_id
- target_angle_deg
- target_x
- target_y

- rotation_direction = clockwise
- rotation_deg = 60

- reported_number
- reported_side
- reported_angle_deg
- report_registered_timestamp
- experimenter_space_timestamp

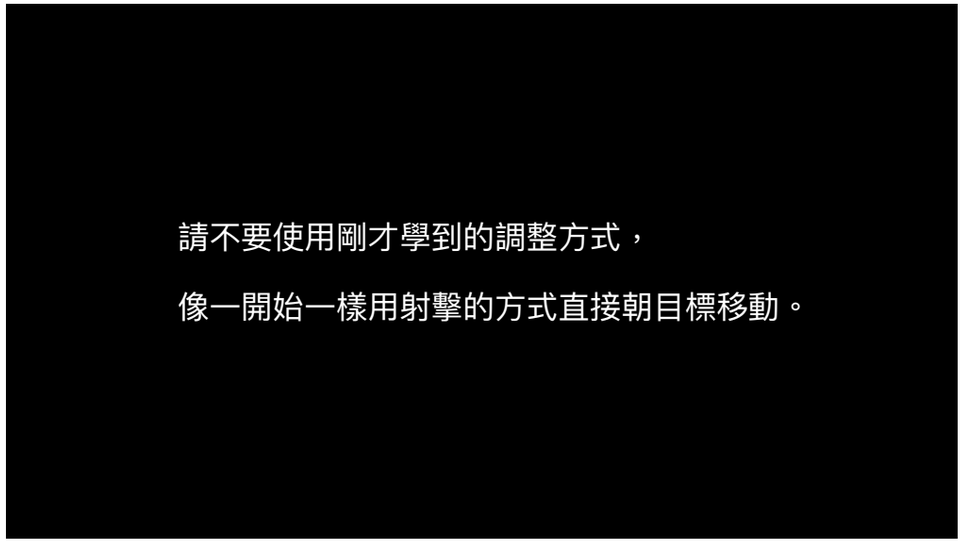
- target_onset_timestamp
- go_signal_timestamp
- movement_onset_timestamp
- movement_end_timestamp

- raw_x
- raw_y
- sample_timestamp
  (record every available input event; do not restrict recording to display frames)

- display_x
- display_y

- hit_target
- boundary_crossed
- timeout

Adaptation- return to baseline trial



Instruction:
- Presented before each Return-to-baseline probe block
- 「請不要使用剛才學到的調整方式，像一開始一樣直接朝目標移動。」
- Press SPACE to start the section

Start layout:
- Home visible at screen center, green
- Cursor visible at home
- Target initially hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Target selection:
- One target is selected pseudorandomly from the 4 possible target locations
- target_angles_deg = [0, 90, 180, 270]
- Across the 4 Return-to-baseline trials, each target appears once

Go signal:
- Participant stays within home for 500 ms

- One target appears, red
- Audio beep
- Movement is enabled
- Home and cursor disappear at movement onset

Movement display:
- Target remains visible
- Cursor NOT visible
- Cursor trajectory NOT visible
- Home hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Visual feedback:
- No visual feedback of hand / touchpad movement
- No 60° rotated cursor is displayed
- No veridical cursor is displayed
- Raw hand / touchpad movement is still recorded normally

End condition:
- Raw movement enters the target zone
- Raw movement crosses workspace boundary (391 px from home)
- Target timeout = 2000 ms after movement is enabled

Feedback:
- Hit + movement time 400–600 ms
  → target turns green + hit bell
- Hit + movement time <400 ms
  → target turns yellow
- Hit + movement time >600 ms
  → target turns blue
- Workspace boundary exceeded / timeout
  → target disappears + miss buzz

Data to record:
- trial_type = adaptation_return_to_baseline
- adaptation_block
- probe_number
- trial_index

- target_id
- target_angle_deg

- target_x
- target_y

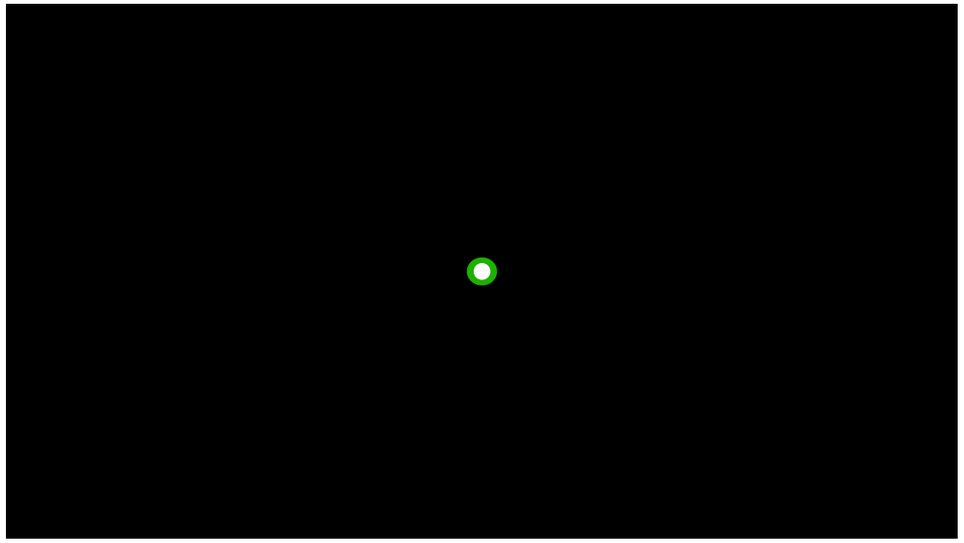
- visual_feedback = none
- rotation_deg = null

- target_onset_timestamp
- go_signal_timestamp
- movement_onset_timestamp
- movement_end_timestamp

- raw_x
- raw_y
- sample_timestamp
  (record every available input event; do not restrict recording to display frames)

- hit_target
- boundary_crossed
- timeout

Adaptation- return to baseline trial 起始



Adaptation – Return-to-baseline Trial

Instruction:
- Presented before each Return-to-baseline probe block
- 「請不要使用剛才學到的調整方式，像一開始一樣直接朝目標移動。」
- Press SPACE to start the section

Start layout:
- Home visible at screen center, green
- Cursor visible at home
- Target initially hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Target selection:
- One target is selected pseudorandomly from the 4 possible target locations
- target_angles_deg = [0, 90, 180, 270]
- Across the 4 Return-to-baseline trials, each target appears once

Go signal:

- Participant stays within home for 500 ms
- One target appears, red
- Audio beep
- Movement is enabled
- Home and cursor disappear at movement onset

Movement display:
- Target remains visible
- Cursor NOT visible
- Cursor trajectory NOT visible
- Home hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Visual feedback:
- No visual feedback of hand / touchpad movement
- No 60° rotated cursor is displayed
- No veridical cursor is displayed
- Raw hand / touchpad movement is still recorded normally

End condition:
- Raw movement enters the target zone
- Raw movement crosses workspace boundary (391 px from home)
- Target timeout = 2000 ms after movement is enabled

Feedback:
- Hit + movement time 400–600 ms
  → target turns green + hit bell
- Hit + movement time <400 ms
  → target turns yellow
- Hit + movement time >600 ms
  → target turns blue
- Workspace boundary exceeded / timeout
  → target disappears + miss buzz

Data to record:
- trial_type = adaptation_return_to_baseline
- adaptation_block
- probe_number
- trial_index

- target_id

- target_angle_deg
- target_x
- target_y

- visual_feedback = none
- rotation_deg = null

- target_onset_timestamp
- go_signal_timestamp
- movement_onset_timestamp
- movement_end_timestamp

- raw_x
- raw_y
- sample_timestamp
  (record every available input event; do not restrict recording to display frames)

- hit_target
- boundary_crossed
- timeout

Adaptation- return to baseline trial 移動



Instruction:
- Presented before each Return-to-baseline probe block
- 「請不要使用剛才學到的調整方式，像一開始一樣直接朝目標移動。」
- Press SPACE to start the section

Start layout:
- Home visible at screen center, green
- Cursor visible at home
- Target initially hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Target selection:
- One target is selected pseudorandomly from the 4 possible target locations
- target_angles_deg = [0, 90, 180, 270]
- Across the 4 Return-to-baseline trials, each target appears once

Go signal:
- Participant stays within home for 500 ms
- One target appears, red

- Audio beep
- Movement is enabled
- Home and cursor disappear at movement onset

Movement display:
- Target remains visible
- Cursor NOT visible
- Cursor trajectory NOT visible
- Home hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Visual feedback:
- No visual feedback of hand / touchpad movement
- No 60° rotated cursor is displayed
- No veridical cursor is displayed
- Raw hand / touchpad movement is still recorded normally

End condition:
- Raw movement enters the target zone
- Raw movement crosses workspace boundary (391 px from home)
- Target timeout = 2000 ms after movement is enabled

Feedback:
- Hit + movement time 400–600 ms
  → target turns green + hit bell
- Hit + movement time <400 ms
  → target turns yellow
- Hit + movement time >600 ms
  → target turns blue
- Workspace boundary exceeded / timeout
  → target disappears + miss buzz

Data to record:
- trial_type = adaptation_return_to_baseline
- adaptation_block
- probe_number
- trial_index

- target_id
- target_angle_deg
- target_x

- target_y

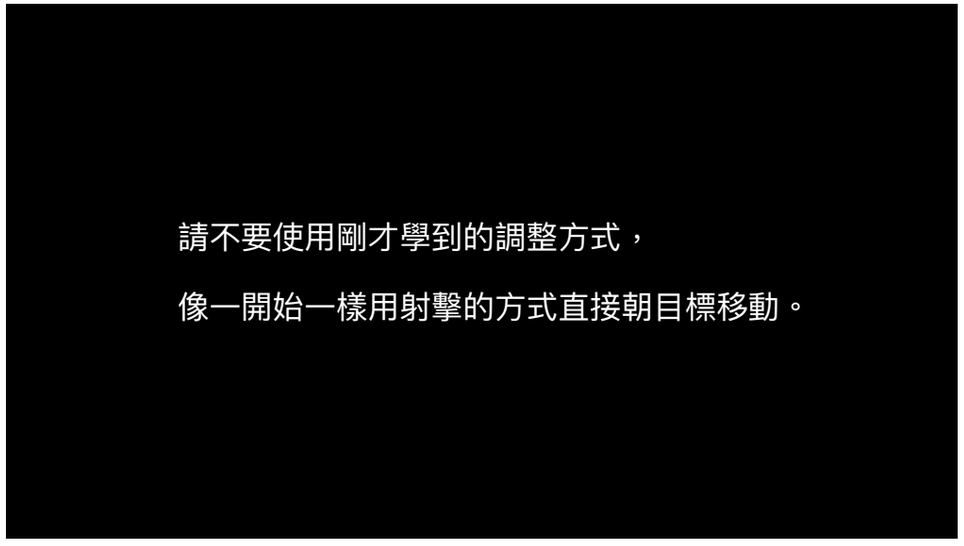
- visual_feedback = none
- rotation_deg = null

- target_onset_timestamp
- go_signal_timestamp
- movement_onset_timestamp
- movement_end_timestamp

- raw_x
- raw_y
- sample_timestamp
  (record every available input event; do not restrict recording to display frames)

- hit_target
- boundary_crossed
- timeout

Post-adaptation- return to baseline trial(section start)



Instruction:
- Presented before each Return-to-baseline probe block
- 「請不要使用剛才學到的調整方式，像一開始一樣直接朝目標移動。」
- Press SPACE to start the section

Start layout:
- Home visible at screen center, green
- Cursor visible at home
- Target initially hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Target selection:
- One target is selected pseudorandomly from the 4 possible target locations
- target_angles_deg = [0, 90, 180, 270]
- Across the 4 Return-to-baseline trials, each target appears once

Go signal:
- Participant stays within home for 500 ms
- One target appears, red

- Audio beep
- Movement is enabled
- Home and cursor disappear at movement onset

Movement display:
- Target remains visible
- Cursor NOT visible
- Cursor trajectory NOT visible
- Home hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Visual feedback:
- No visual feedback of hand / touchpad movement
- No 60° rotated cursor is displayed
- No veridical cursor is displayed
- Raw hand / touchpad movement is still recorded normally

End condition:
- Raw movement enters the target zone
- Raw movement crosses workspace boundary (391 px from home)
- Target timeout = 2000 ms after movement is enabled

Feedback:
- Hit + movement time 400–600 ms
  → target turns green + hit bell
- Hit + movement time <400 ms
  → target turns yellow
- Hit + movement time >600 ms
  → target turns blue
- Workspace boundary exceeded / timeout
  → target disappears + miss buzz

Data to record:
- trial_type = adaptation_return_to_baseline
- adaptation_block
- probe_number
- trial_index

- target_id
- target_angle_deg
- target_x

- target_y

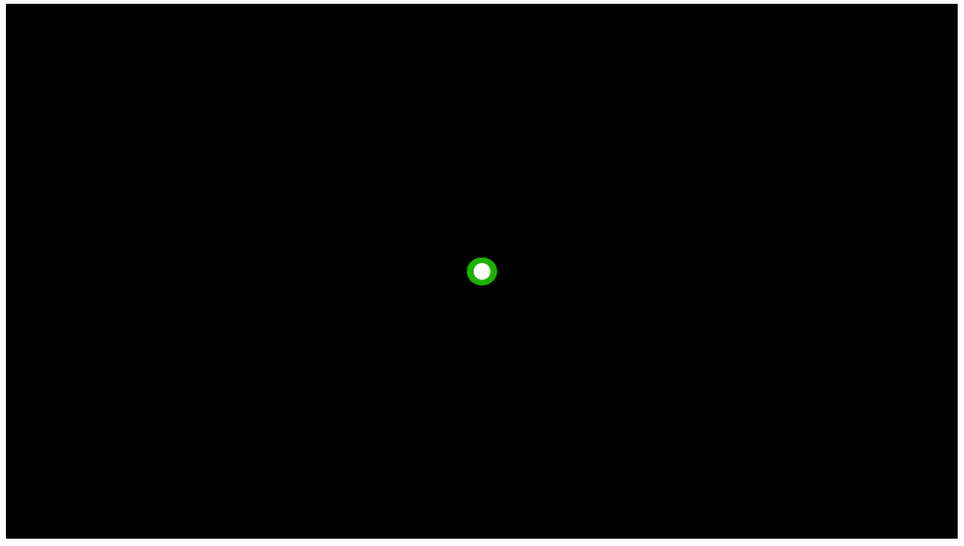
- visual_feedback = none
- rotation_deg = null

- target_onset_timestamp
- go_signal_timestamp
- movement_onset_timestamp
- movement_end_timestamp

- raw_x
- raw_y
- sample_timestamp
  (record every available input event; do not restrict recording to display frames)

- hit_target
- boundary_crossed
- timeout

Post-adaptation- return to baseline trial (起始)



Instruction:
- Presented before each Return-to-baseline probe block
- 「請不要使用剛才學到的調整方式，像一開始一樣直接朝目標移動。」
- Press SPACE to start the section

Start layout:
- Home visible at screen center, green
- Cursor visible at home
- Target initially hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Target selection:
- One target is selected pseudorandomly from the 4 possible target locations
- target_angles_deg = [0, 90, 180, 270]
- Across the 4 Return-to-baseline trials, each target appears once

Go signal:
- Participant stays within home for 500 ms
- One target appears, red

- Audio beep
- Movement is enabled
- Home and cursor disappear at movement onset

Movement display:
- Target remains visible
- Cursor NOT visible
- Cursor trajectory NOT visible
- Home hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Visual feedback:
- No visual feedback of hand / touchpad movement
- No 60° rotated cursor is displayed
- No veridical cursor is displayed
- Raw hand / touchpad movement is still recorded normally

End condition:
- Raw movement enters the target zone
- Raw movement crosses workspace boundary (391 px from home)
- Target timeout = 2000 ms after movement is enabled

Feedback:
- Hit + movement time 400–600 ms
  → target turns green + hit bell
- Hit + movement time <400 ms
  → target turns yellow
- Hit + movement time >600 ms
  → target turns blue
- Workspace boundary exceeded / timeout
  → target disappears + miss buzz

Data to record:
- trial_type = adaptation_return_to_baseline
- adaptation_block
- probe_number
- trial_index

- target_id
- target_angle_deg
- target_x

- target_y

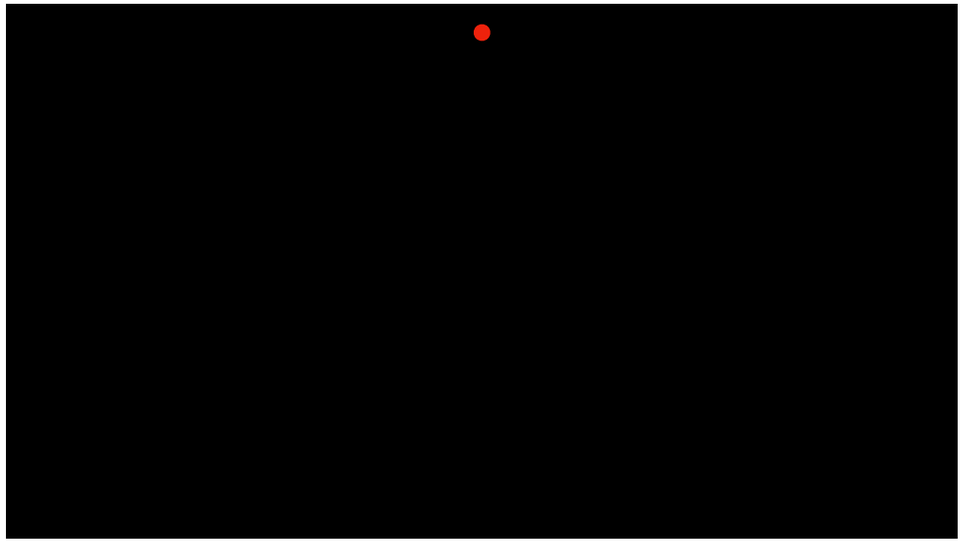
- visual_feedback = none
- rotation_deg = null

- target_onset_timestamp
- go_signal_timestamp
- movement_onset_timestamp
- movement_end_timestamp

- raw_x
- raw_y
- sample_timestamp
  (record every available input event; do not restrict recording to display frames)

- hit_target
- boundary_crossed
- timeout

Adaptation- return to baseline trial 起始



Instruction:
- Presented before each Return-to-baseline probe block
- 「請不要使用剛才學到的調整方式，像一開始一樣直接朝目標移動。」
- Press SPACE to start the section

Start layout:
- Home visible at screen center, green
- Cursor visible at home
- Target initially hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Target selection:
- One target is selected pseudorandomly from the 4 possible target locations
- target_angles_deg = [0, 90, 180, 270]
- Across the 4 Return-to-baseline trials, each target appears once

Go signal:
- Participant stays within home for 500 ms
- One target appears, red

- Audio beep
- Movement is enabled
- Home and cursor disappear at movement onset

Movement display:
- Target remains visible
- Cursor NOT visible
- Cursor trajectory NOT visible
- Home hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Visual feedback:
- No visual feedback of hand / touchpad movement
- No 60° rotated cursor is displayed
- No veridical cursor is displayed
- Raw hand / touchpad movement is still recorded normally

End condition:
- Raw movement enters the target zone
- Raw movement crosses workspace boundary (391 px from home)
- Target timeout = 2000 ms after movement is enabled

Feedback:
- Hit + movement time 400–600 ms
  → target turns green + hit bell
- Hit + movement time <400 ms
  → target turns yellow
- Hit + movement time >600 ms
  → target turns blue
- Workspace boundary exceeded / timeout
  → target disappears + miss buzz

Data to record:
- trial_type = adaptation_return_to_baseline
- adaptation_block
- probe_number
- trial_index

- target_id
- target_angle_deg
- target_x

- target_y

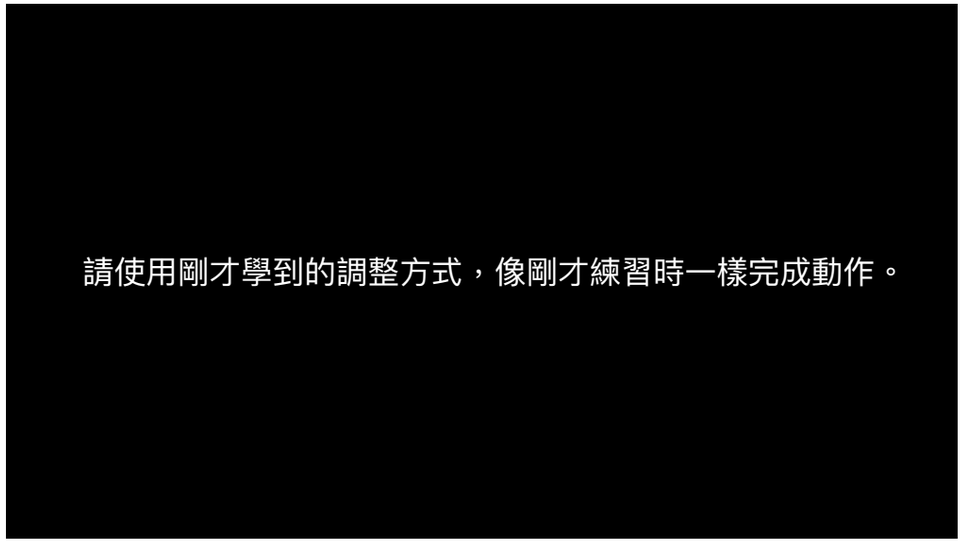
- visual_feedback = none
- rotation_deg = null

- target_onset_timestamp
- go_signal_timestamp
- movement_onset_timestamp
- movement_end_timestamp

- raw_x
- raw_y
- sample_timestamp
  (record every available input event; do not restrict recording to display frames)

- hit_target
- boundary_crossed
- timeout

Post-adaptation- use learned adjust trial(section start)



Instruction:
- Presented before each Return-to-baseline probe block
- 「請使用剛才學到的調整方式，像剛才練習時一樣完成動作。」
- Press SPACE to start the section

Start layout:
- Home visible at screen center, green
- Cursor visible at home
- Target initially hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Target selection:
- One target is selected pseudorandomly from the 4 possible target locations
- target_angles_deg = [0, 90, 180, 270]
- Across the 4 Return-to-baseline trials, each target appears once

Go signal:
- Participant stays within home for 500 ms
- One target appears, red

- Audio beep
- Movement is enabled
- Home and cursor disappear at movement onset

Movement display:
- Target remains visible
- Cursor NOT visible
- Cursor trajectory NOT visible
- Home hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Visual feedback:
- No visual feedback of hand / touchpad movement
- No 60° rotated cursor is displayed
- No veridical cursor is displayed
- Raw hand / touchpad movement is still recorded normally

End condition:
- Raw movement enters the target zone
- Raw movement crosses workspace boundary (391 px from home)
- Target timeout = 2000 ms after movement is enabled

Feedback:
- Hit + movement time 400–600 ms
  → target turns green + hit bell
- Hit + movement time <400 ms
  → target turns yellow
- Hit + movement time >600 ms
  → target turns blue
- Workspace boundary exceeded / timeout
  → target disappears + miss buzz

Data to record:
- trial_type = adaptation_return_to_baseline
- adaptation_block
- probe_number
- trial_index

- target_id
- target_angle_deg
- target_x

- target_y

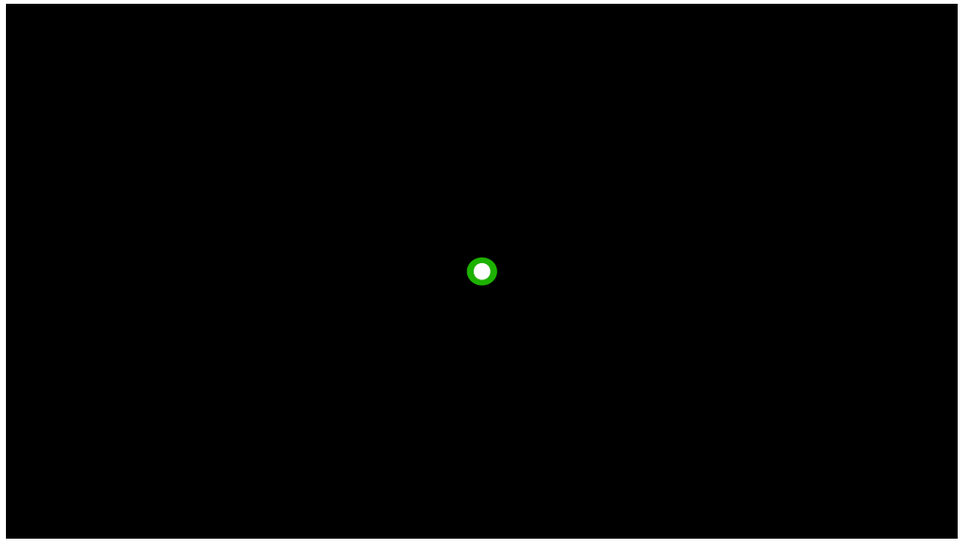
- visual_feedback = none
- rotation_deg = null

- target_onset_timestamp
- go_signal_timestamp
- movement_onset_timestamp
- movement_end_timestamp

- raw_x
- raw_y
- sample_timestamp
  (record every available input event; do not restrict recording to display frames)

- hit_target
- boundary_crossed
- timeout

Post-adaptation- use learned adjust trial (起始)



Instruction:
- Presented before each Return-to-baseline probe block
- 「請使用剛才學到的調整方式，像剛才練習時一樣完成動作。」
- Press SPACE to start the section

Start layout:
- Home visible at screen center, green
- Cursor visible at home
- Target initially hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Target selection:
- One target is selected pseudorandomly from the 4 possible target locations
- target_angles_deg = [0, 90, 180, 270]
- Across the 4 Return-to-baseline trials, each target appears once

Go signal:
- Participant stays within home for 500 ms
- One target appears, red

- Audio beep
- Movement is enabled
- Home and cursor disappear at movement onset

Movement display:
- Target remains visible
- Cursor NOT visible
- Cursor trajectory NOT visible
- Home hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Visual feedback:
- No visual feedback of hand / touchpad movement
- No 60° rotated cursor is displayed
- No veridical cursor is displayed
- Raw hand / touchpad movement is still recorded normally

End condition:
- Raw movement enters the target zone
- Raw movement crosses workspace boundary (391 px from home)
- Target timeout = 2000 ms after movement is enabled

Feedback:
- Hit + movement time 400–600 ms
  → target turns green + hit bell
- Hit + movement time <400 ms
  → target turns yellow
- Hit + movement time >600 ms
  → target turns blue
- Workspace boundary exceeded / timeout
  → target disappears + miss buzz

Data to record:
- trial_type = adaptation_return_to_baseline
- adaptation_block
- probe_number
- trial_index

- target_id
- target_angle_deg
- target_x

- target_y

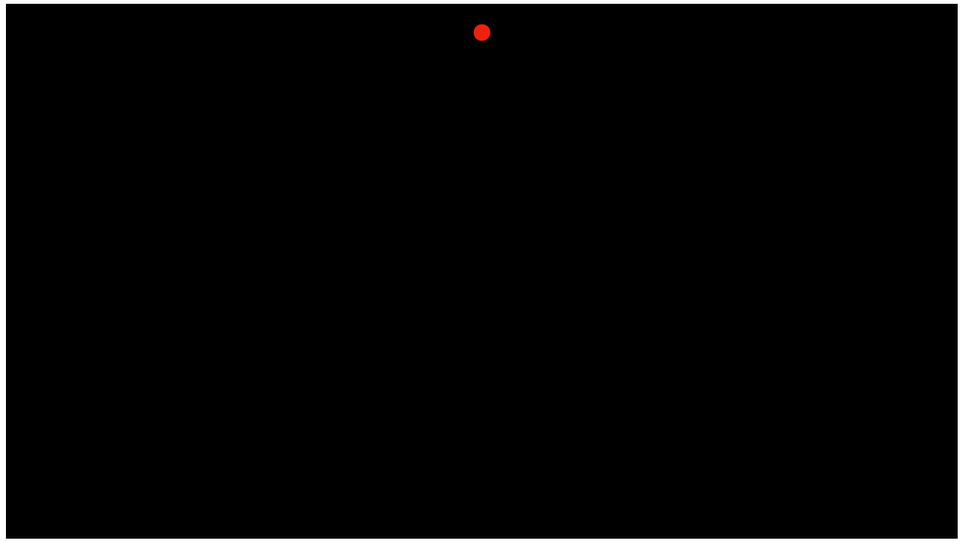
- visual_feedback = none
- rotation_deg = null

- target_onset_timestamp
- go_signal_timestamp
- movement_onset_timestamp
- movement_end_timestamp

- raw_x
- raw_y
- sample_timestamp
  (record every available input event; do not restrict recording to display frames)

- hit_target
- boundary_crossed
- timeout

post-adaptation- use learned adjust trial （過程）



Instruction:
- Presented before each Return-to-baseline probe block
- 「請使用剛才學到的調整方式，像剛才練習時一樣完成動作。」
- Press SPACE to start the section

Start layout:
- Home visible at screen center, green
- Cursor visible at home
- Target initially hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Target selection:
- One target is selected pseudorandomly from the 4 possible target locations
- target_angles_deg = [0, 90, 180, 270]
- Across the 4 Return-to-baseline trials, each target appears once

Go signal:
- Participant stays within home for 500 ms
- One target appears, red

- Audio beep
- Movement is enabled
- Home and cursor disappear at movement onset

Movement display:
- Target remains visible
- Cursor NOT visible
- Cursor trajectory NOT visible
- Home hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Visual feedback:
- No visual feedback of hand / touchpad movement
- No 60° rotated cursor is displayed
- Raw hand / touchpad movement is still recorded normally

End condition:
- Raw movement enters the target zone
- Raw movement crosses workspace boundary (391 px from home)
- Target timeout = 2000 ms after movement is enabled

Feedback:
- Hit + movement time 400–600 ms
  → target turns green + hit bell
- Hit + movement time <400 ms
  → target turns yellow
- Hit + movement time >600 ms
  → target turns blue
- Workspace boundary exceeded / timeout
  → target disappears + miss buzz

Data to record:
- trial_type = adaptation_return_to_baseline
- adaptation_block
- probe_number
- trial_index

- target_id
- target_angle_deg
- target_x
- target_y

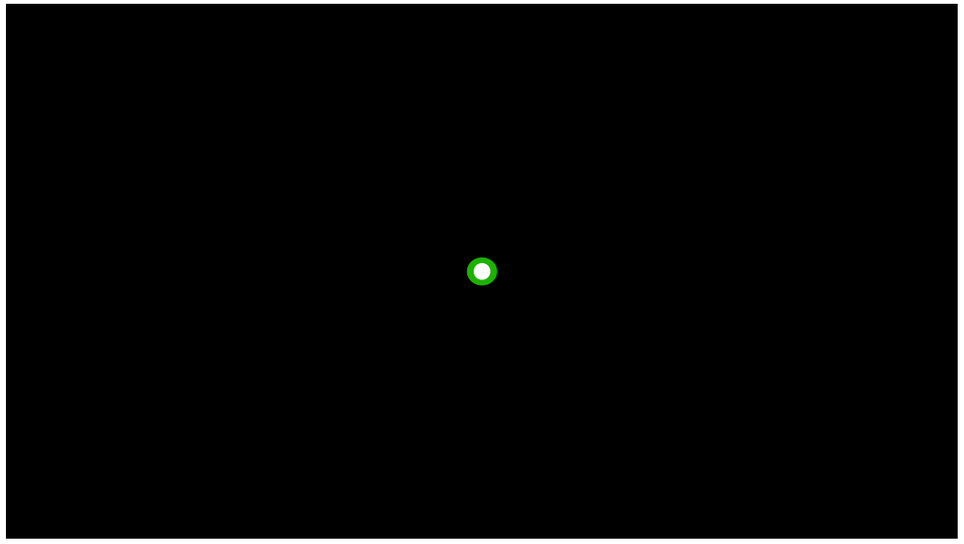
- visual_feedback = none
- rotation_deg = null

- target_onset_timestamp
- go_signal_timestamp
- movement_onset_timestamp
- movement_end_timestamp

- raw_x
- raw_y
- sample_timestamp
  (record every available input event; do not restrict recording to display frames)

- hit_target
- boundary_crossed
- timeout

Post-adaptation- re-adaptation trial （起始）



Start layout:
- Home visible at screen center, green
- Cursor visible at home
- Target initially hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Target selection:
- One target is selected pseudorandomly from the 4 possible target locations
- target_angles_deg = [0, 90, 180, 270]
- Only the selected target is visible to the participant

Go signal:
- Participant stays within home for 500 ms
- One target appears, red
- Audio beep
- Movement is enabled
- Home disappears at movement onset

Movement display:

- Target remains visible
- Rotated cursor visible continuously during movement
- Home hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Cursor mapping:
- Fixed visuomotor rotation
- rotation_deg = 60
- Same rotation magnitude and direction throughout the adaptation phase
- Rotation is applied to the displayed cursor only; raw movement is recorded separately

End condition:
- Display cursor enters target zone
- Raw movement crosses workspace boundary (391 px from home)
- Target timeout = 2000 ms

Feedback:
- Hit + movement time 400–600 ms
  → target turns green + hit bell
- Hit + movement time <400 ms
  → target turns yellow
- Hit + movement time >600 ms
  → target turns blue
- Workspace boundary exceeded / timeout
  → target disappears + miss buzz

Data to record:
- trial_type = adaptation_regular
- adaptation_block
- trial_index

- target_id
- target_angle_deg
- target_x
- target_y

- rotation_deg

- target_onset_timestamp
- go_signal_timestamp
- movement_onset_timestamp

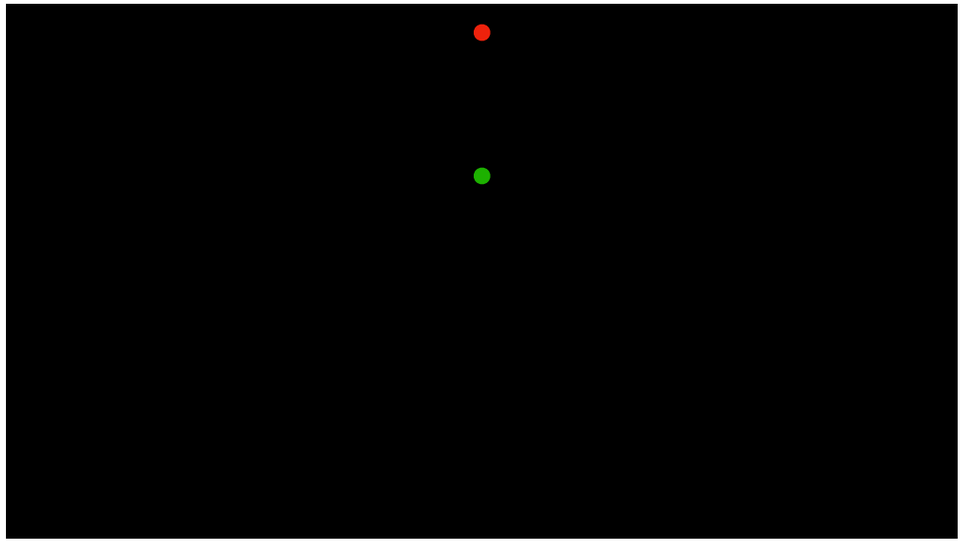
- movement_end_timestamp

- raw_x
- raw_y
- sample_timestamp
  (record every available input event; do not restrict recording to display frames)

- display_x
- display_y

- hit_target
- boundary_crossed
- timeout

Post-adaptation- re-adaptation trial (過程)



Start layout:
- Home visible at screen center, green
- Cursor visible at home
- Target initially hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Target selection:
- One target is selected pseudorandomly from the 4 possible target locations
- target_angles_deg = [0, 90, 180, 270]
- Only the selected target is visible to the participant

Go signal:
- Participant stays within home for 500 ms
- One target appears, red
- Audio beep
- Movement is enabled
- Home disappears at movement onset

Movement display:

- Target remains visible
- Rotated cursor visible continuously during movement
- Home hidden
- Numbered landmarks hidden
- Workspace boundary hidden

Cursor mapping:
- Fixed visuomotor rotation
- rotation_deg = 60
- Same rotation magnitude and direction throughout the adaptation phase
- Rotation is applied to the displayed cursor only; raw movement is recorded separately

End condition:
- Display cursor enters target zone
- Raw movement crosses workspace boundary (391 px from home)
- Target timeout = 2000 ms

Feedback:
- Hit + movement time 400–600 ms
  → target turns green + hit bell
- Hit + movement time <400 ms
  → target turns yellow
- Hit + movement time >600 ms
  → target turns blue
- Workspace boundary exceeded / timeout
  → target disappears + miss buzz

Data to record:
- trial_type = adaptation_regular
- adaptation_block
- trial_index

- target_id
- target_angle_deg
- target_x
- target_y

- rotation_deg

- target_onset_timestamp
- go_signal_timestamp
- movement_onset_timestamp

- movement_end_timestamp

- raw_x
- raw_y
- sample_timestamp
  (record every available input event; do not restrict recording to display frames)

- display_x
- display_y

- hit_target
- boundary_crossed
- timeout